NEWTON'S LAWS OF MOTION - Revised Core 1

Build the force-and-momentum reasoning system before transfer

REVISED CORE 1

Core 1 intent: never start with a formula name. Start by choosing the body or system, identify the interactions/state, build the representation, then select the law. Every reasoning move used in Core 2 must already be taught here.

1 CHOOSE	Which body or system is tracked?
2 REPRESENT	FBD, interaction map, momentum state, graph, or system boundary.
3 CONNECT	Newton I / $\Sigma F=ma$ / $\Delta p/\Delta t$ / Newton III / momentum conservation.
4 EXECUTE + CHECK	Keep signs, units, bodies and system membership explicit.

Competency map

Atomic skills that Core 2 is allowed to use

REVISED CORE 1

01	**Motion is not force**	Constant velocity can coexist with zero net force
02	**Choose the body before drawing forces**	Interactions become forces only after the body is chosen
03	**Net force is a vector sum**	Balanced forces and no forces are different statements
04	**Newton II and scaling**	Acceleration responds to net force and mass together
05	**Stopping force needs two models**	Kinematics supplies acceleration; Newton II supplies force
06	**Momentum is a signed state**	Momentum carries the direction of velocity
07	**Impulse and average force**	The same momentum change can be spread over more time
08	**Newton III pairs**	Equal and opposite does not mean same body
09	**Momentum conservation is a system statement**	Choose the system and check external impulse first
10	**Recoil, rockets and exchanged momentum**	Equal momentum can mean very different speeds
11	**Read force from a motion graph**	v-t slope → acceleration → force

NLM 01 - Motion is not force

Constant velocity can coexist with zero net force

LEARN



WHAT IS HAPPENING?

Newton I links change of velocity to net external force. Motion itself does not require a sustaining net force.

KEY IDEA

If velocity is constant in an inertial frame, acceleration is zero and therefore the net external force is zero.

HOW IT WORKS

1. Choose the body and ask whether its velocity is changing.
2. Draw all external forces on that body.
3. Add them vectorially before discussing motion.
4. Check: zero net force means constant velocity, including rest.

Practice 01 - Motion is not force

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A crate moves right at 3 m/s while 60 N right and 60 N left act on it. $F_{net}=0$, so a=0; it continues at constant velocity.

GUIDED TRY

A puck glides at constant velocity on an ideal frictionless surface. What horizontal net force is required?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

A moving object does not automatically need a forward net force.

QUICK CHECK

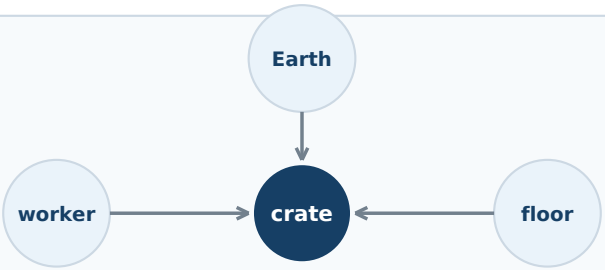
A puck glides at constant velocity on an ideal frictionless surface. What horizontal net force is required?

Answer: 0 N.

NLM 02 - Choose the body before drawing forces

Interactions become forces only after the body is chosen

LEARN



Interactions first; forces on the chosen body second.

WHAT IS HAPPENING?

A free-body diagram is a diagram of one chosen body. First identify what interacts with it; then draw only the forces acting on that body.

KEY IDEA

The phrase “force on X by Y” prevents body confusion and Newton-III mistakes.

HOW IT WORKS

1. Circle the body/system being analysed.
2. List interacting bodies: Earth, surface, rope, hand, etc.
3. Convert each interaction into one force arrow on the chosen body.
4. Do not include forces the chosen body exerts on something else.

Practice 02 - Choose the body before drawing forces

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

For a book on a table, the book FBD contains Earth-on-book weight and table-on-book normal. The book-on-table force belongs to the table FBD.

GUIDED TRY

A crate is pushed by a worker while sliding on a floor. Name three possible external interactions on the crate.

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

An interaction pair is not automatically two arrows on one FBD.

QUICK CHECK

A crate is pushed by a worker while sliding on a floor. Name three possible external interactions on the crate.

Answer: Worker-crate, Earth-crate, floor-crate.

NLM 03 - Net force is a vector sum

Balanced forces and no forces are different statements

LEARN



WHAT IS HAPPENING?

Newton II uses the net external force, not whichever force is easiest to notice. Balanced forces can be nonzero individually.

KEY IDEA

Choose a positive axis, assign signs to force components, and sum them.

HOW IT WORKS

1. Choose +x and +y.
2. Resolve forces into components when needed.
3. Compute ΣF_x and ΣF_y.
4. Only then infer acceleration components.

Practice 03 - Net force is a vector sum

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A cart has 25 N right and 10 N left. With right positive, $F_{net,x}=25-10=15$ N right.

GUIDED TRY

A box has 30 N right and 30 N left. What is its horizontal acceleration?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Do not confuse "$F_{net}=0$" with "there are no forces."

QUICK CHECK

A box has 30 N right and 30 N left. What is its horizontal acceleration?

Answer: 0 m/s², regardless of its current velocity.

NLM 04 - Newton II and scaling

Acceleration responds to net force and mass together

LEARN

F	m	a
60 N	5 kg	12 m/s²
60 N	10 kg	6 m/s²
30 N	5 kg	6 m/s²

WHAT IS HAPPENING?

For a chosen body of constant mass, $F_{net}=ma$. At fixed mass, a scales with net force; at fixed force, a scales inversely with mass.

KEY IDEA

For changes in both force and mass, compare ratios before substituting numbers.

HOW IT WORKS

1. Write $a_1=F_1/m_1$ and $a_2=F_2/m_2$.
2. Form $a_2/a_1=(F_2/F_1)(m_1/m_2)$.
3. Keep direction from the signed net force.
4. Check units: 1 N = 1 kg·m/s².

Practice 04 - Newton II and scaling

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

If force halves while mass doubles, a becomes (1/2)(1/2)=1/4 of its original value.

GUIDED TRY

A 40 N net force accelerates 5 kg. What is a?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

“Half the force means half the acceleration” is true only if mass is unchanged.

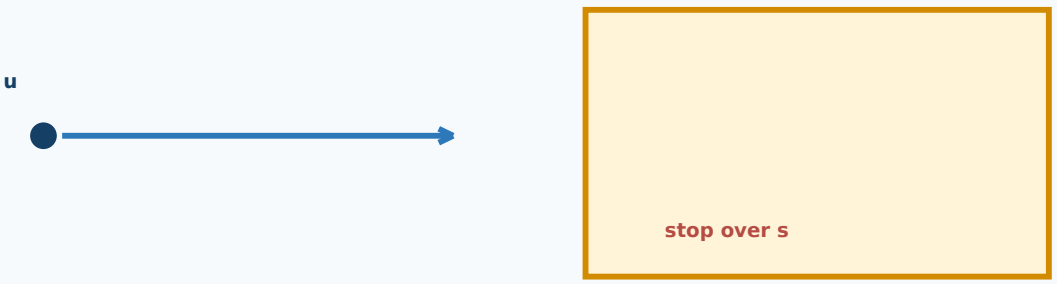
QUICK CHECK

A 40 N net force accelerates 5 kg. What is a? **Answer: 8 m/s².**

NLM 05 - Stopping force needs two models

Kinematics supplies acceleration; Newton II supplies force

LEARN



WHAT IS HAPPENING?

Stopping-over-distance problems often require a kinematics step before a force step. Do not jump directly from distance to force.

KEY IDEA

For constant stopping acceleration, find a from $v^2=u^2+2as$, then use $F_{net}=ma$; find time separately if requested.

HOW IT WORKS

1. Write the motion state u, v=0, s.
2. Use $v^2=u^2+2as$ to find a.
3. Use F=ma with sign/direction.
4. If time is needed, use v=u+at after a is known.

Practice 05 - Stopping force needs two models

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A 0.020 kg pellet at 100 m/s stops uniformly in 0.10 m. a=(0-10000)/(0.20)=-50000 m/s², so F=-1000 N.

GUIDED TRY

If the same pellet stops over twice the distance, what happens to |a| and |F|?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Do not mix the kinematic distance relation with Newton II as though they were one formula.

QUICK CHECK

If the same pellet stops over twice the distance, what happens to |a| and |F|?

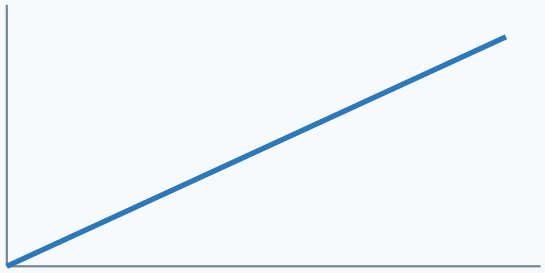
Answer: Both halve.

NLM 06 - Momentum is a signed state

Momentum carries the direction of velocity

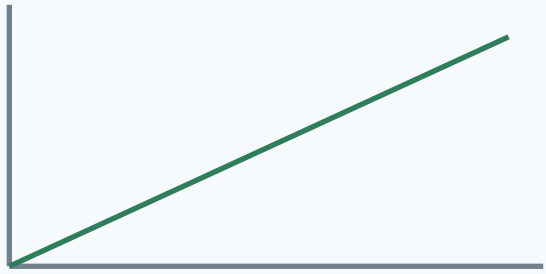
LEARN

p vs v



slope = mass

p vs m



slope = velocity

WHAT IS HAPPENING?

Momentum is p=mv. In one dimension its sign follows velocity. A change in direction can make Δp much larger than a simple speed difference.

KEY IDEA

Always write p_i, p_f, then $\Delta p = p_f - p_i$.

HOW IT WORKS

1. Choose a positive direction.
2. Write signed velocities.
3. Convert to signed momenta.
4. Subtract final minus initial.

Practice 06 - Momentum is a signed state

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A 0.20 kg ball changes from +8 m/s to -6 m/s. $\Delta p = 0.20(-6) - 0.20(8) = -2.8$ kg·m/s.

GUIDED TRY

At fixed mass, what is the shape of a p-v graph?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Do not discard signs when a body rebounds.

QUICK CHECK

At fixed mass, what is the shape of a p-v graph?

Answer: A straight line through the origin; slope = mass.

NLM 07 - Impulse and average force

The same momentum change can be spread over more time

LEARN

same Δp, different stopping time



WHAT IS HAPPENING?

Average force measures how quickly momentum changes: $F_{avg}=\Delta p/\Delta t$. Increasing stopping time reduces force magnitude for the same Δp.

KEY IDEA

This is the physics behind moving the hands backward while catching.

HOW IT WORKS

1. Find the signed Δp.
2. Identify the interaction duration Δt.
3. Divide Δp by Δt.
4. Check force direction against Δp, not against the initial velocity alone.

Practice 07 - Impulse and average force

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A 0.50 kg ball at +12 m/s stops in 0.30 s: Δp=-6 kg·m/s, F_{avg}=-20 N.

GUIDED TRY

If the same ball is stopped in twice the time, what happens to average force magnitude?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Momentum and force are not the same quantity.

QUICK CHECK

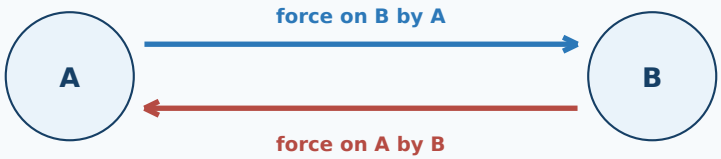
If the same ball is stopped in twice the time, what happens to average force magnitude?

Answer: It halves.

NLM 08 - Newton III pairs

Equal and opposite does not mean same body

LEARN



WHAT IS HAPPENING?

A third-law pair belongs to the same interaction, has equal magnitude and opposite direction, and acts on different bodies.

KEY IDEA

Use complete names: force on B by A; force on A by B.

HOW IT WORKS

1. Name both interacting bodies.
2. Write the two force names.
3. Place each force on the correct body’s FBD.
4. Use Newton II separately to compare accelerations.

Practice 08 - Newton III pairs

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

Earth pulls an apple downward with the same force magnitude that the apple pulls Earth upward. Their accelerations differ because their masses differ.

GUIDED TRY

What is the third-law partner of the table-on-book normal force?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Weight and normal on a resting book are balanced forces on one body, not a Newton-III pair.

QUICK CHECK

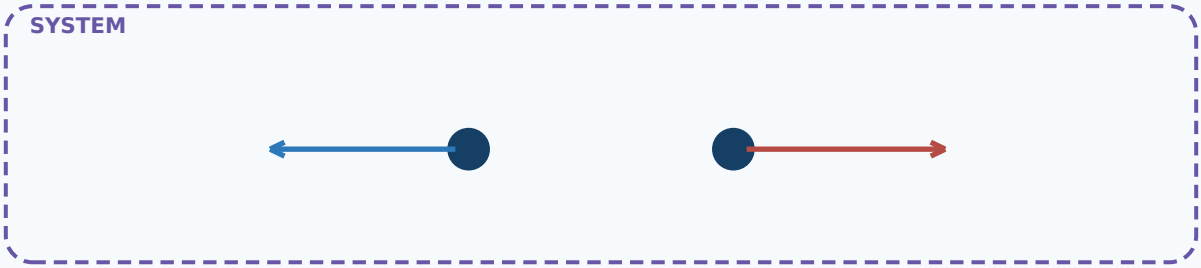
What is the third-law partner of the table-on-book normal force?

Answer: The book-on-table contact force.

NLM 09 - Momentum conservation is a system statement

Choose the system and check external impulse first

LEARN



Check external impulse before conserving total momentum.

WHAT IS HAPPENING?

Total momentum is conserved only for a chosen system when the relevant external impulse over the interaction is negligible.

KEY IDEA

The equation $p_{before,total}=p_{after,total}$ comes after the system/interval check.

HOW IT WORKS

1. Draw a system boundary.
2. Choose a positive direction.
3. List all system momenta before and after.
4. Check external impulse; then equate totals if valid.

Practice 09 - Momentum conservation is a system statement

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

Two skaters push apart from rest. For the two-skater system, initial total p=0, so their final momenta are equal in magnitude and opposite in direction.

GUIDED TRY

A falling ball speeds up under gravity. Is the ball-only momentum conserved?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Do not claim conservation from one object’s momentum alone when an external force acts on it.

QUICK CHECK

A falling ball speeds up under gravity. Is the ball-only momentum conserved?

Answer: No; gravity gives the ball external impulse.

NLM 10 - Recoil, rockets and exchanged momentum

Equal momentum can mean very different speeds

LEARN



WHAT IS HAPPENING?

In recoil/explosion problems with negligible external impulse, momentum exchanged internally appears with opposite signs.

KEY IDEA

For equal momentum magnitude, speed is inversely proportional to mass.

HOW IT WORKS

1. Choose launcher+projectile (or rocket+exhaust) as the system.
2. Set initial total momentum.
3. Assign opposite signs after separation.
4. Solve one object’s speed from total momentum conservation.

Practice 10 - Recoil, rockets and exchanged momentum

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A 0.020 kg bullet leaves a 2.0 kg pistol at +150 m/s from rest. 0=(0.020)(150)+2v, so v=-1.5 m/s.

GUIDED TRY

Two rifles fire identical bullets with the same bullet momentum. Which rifle has larger recoil speed?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

A lighter rifle recoils faster for the same bullet momentum, not with a larger recoil momentum.

QUICK CHECK

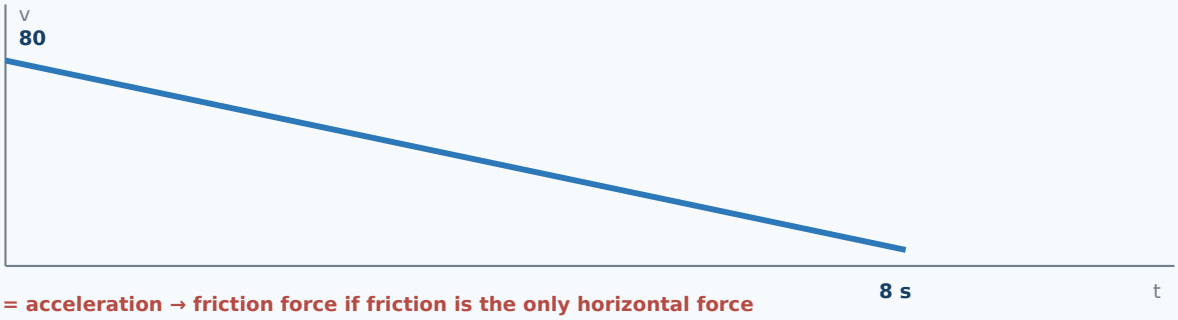
Two rifles fire identical bullets with the same bullet momentum. Which rifle has larger recoil speed?

Answer: The lighter rifle.

NLM 11 - Read force from a motion graph

v-t slope → acceleration → force

LEARN



slope = acceleration → friction force if friction is the only horizontal force

WHAT IS HAPPENING?

A velocity-time graph gives acceleration from its slope. If friction is the only horizontal force, then $F_{friction}=ma$.

KEY IDEA

The graph height is velocity; the slope is acceleration.

HOW IT WORKS

1. Read two points on the straight segment.
2. Compute $a=\Delta v/\Delta t$.
3. Convert mass to kilograms.
4. Apply $F=ma$ and interpret the sign.

Practice 11 - Read force from a motion graph

Worked -> guided -> faded -> contrast

PRACTISE

WORKED EXAMPLE

A 50 g object slows from 80 m/s to 0 in 8 s. $a=-10\ m/s^2$; $F=0.050(-10)=-0.50$ N.

GUIDED TRY

If the same velocity change takes twice as long, what happens to |a| and |F|?

Guide: name the body/system -> draw the representation -> state the law before calculating.

FADED TRY

Change one feature of the worked example (direction, mass, force, time, or system boundary). Which parts of the reasoning remain valid and which must change?

WATCH OUT

Do not use the area under a v-t graph when the question asks for acceleration or force.

QUICK CHECK

If the same velocity change takes twice as long, what happens to |a| and |F|? **Answer: Both halve.**

First-move atlas

Use this after learning; it is not a formula sheet

REVISED CORE 1

WHEN YOU SEE...	FIRST REPRESENTATION	FIRST PHYSICS MOVE
constant velocity / rest	one-body FBD	test whether ΣF=0
force question	choose body + interactions	draw only forces on that body
several forces	signed component sum	find Fnet before ma
force/mass change	old/new state ratio	compare F/m
stopping over distance	kinematic state	find a first, then F=ma
catch / collision time	signed p states	Δp then Δp/Δt
action-reaction	two-body interaction pair	force on B by A ↔ force on A by B
collision / recoil	system boundary	check external impulse, then conserve total p
rocket / rifle	opposite signed momenta	lighter mass → larger speed for same |p|
v-t friction graph	graph slope	a=Δv/Δt then F=ma
p-v or p-m graph	read axes + slope	slope gives m or v